Sound speed and resonance

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use travel time and resonant conditions to interpret sound.

SUCCESS CRITERIA - I can...

1. Account for round-trip distance in an echo.
2. Identify resonance as a response near a natural frequency.
3. Use an appropriate wavelength relationship.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “echo” and “resonance” to explain your model or result. Sentence starters are optional.

echo: a sound received after reflection

resonance: a strong response near a system’s natural frequency

period: time for one complete cycle or orbit

01 Warm-up: bring back an earlier idea

Two waves meet the same opening with wavelengths 3 cm and 5 cm. Which generally diffracts more? **Explain** the relevant comparison.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $d_{echo} = v\,\Delta t/2$; v = 340 m/s.
Echo time is a round trip. Resonance occurs near a natural mode under suitable damping.

Worked reasoning: In an ideal tube open at both ends, the fundamental wavelength is twice its length. For L=0.5 m and sound speed 340 m/s, f=340/1=340 Hz.

Your turn: Sketch the fundamental with displacement antinodes at both ends.

Sentence starter: The … in my model represents … . Its label shows … .

open-open tube

Draw a displacement pattern; label nodes and antinodes.

Marzano check - circle: ① ② ③

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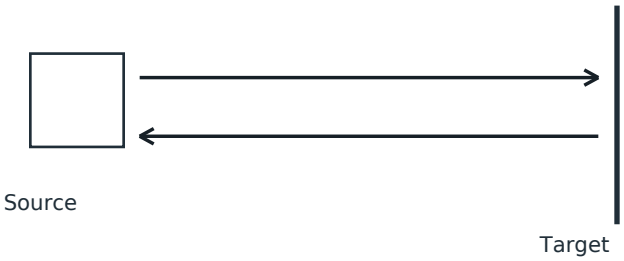
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

An echo returns from a wall.
Draw the outgoing and returning path. Why does sound's total travel distance exceed wall range?

Sentence starter: My claim is … . The diagram/data show …, so … .

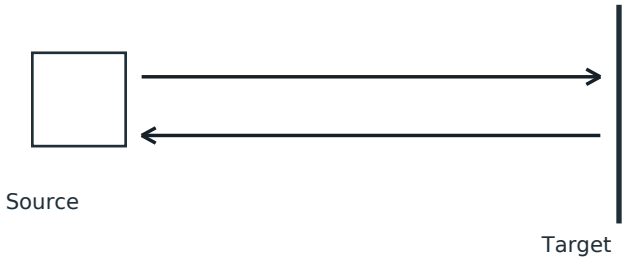


Marzano check - circle: 1 2 3

04 Apply the physics

In a scaled echo simulation, a pulse travels at 10 m/s and returns after 4 s.
Label both legs of the trip.
Calculate the total path distance and the one-way distance to the wall.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: 1 2 3

05 Explain your reasoning

Why does a resonant response not mean that any object can respond strongly to every frequency?

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A heat engine receives 600 J heat and outputs 180 J work per cycle.
Find efficiency and rejected heat.
Label energy entering, leaving and stored.

Word help: efficiency = useful output divided by input

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

07 Exit check

An echo’s round-trip path is 60 m. How far away is the wall?

Sentence starter: I notice … . This supports … because … .



Marzano check - circle: 1 2 3

08 STRETCH: test the model

Return to the worked example in problem 2.
Compare fundamental frequencies for ideal open-open tubes of lengths 0.5 and 1.0 m at the same sound speed.

Word help: ideal = using stated simplifying conditions

Sentence starter: Both … . They differ in … because … .

open-open tube

Draw a displacement pattern; label nodes and antinodes.

Marzano check - circle: 1 2 3